Session 1

Global landscape of data governance institutions and frameworks

Definitions

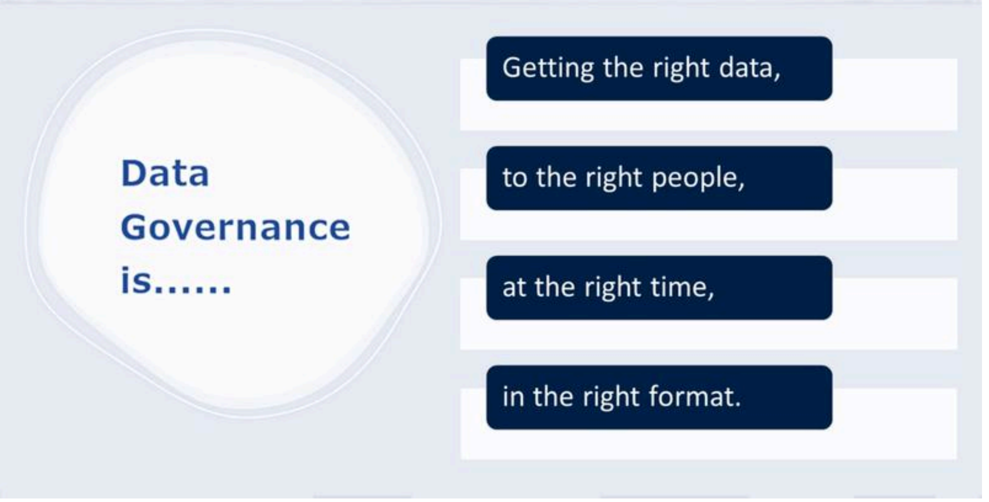
What is data?

“When we refer to data, we mean information about people, things and systems. … Data about people can include personal data, such as basic contact details, records generated through interaction with services or the web, or information about their physical characteristics (biometrics) - and it can also extend to population-level data, such as demographics. Data can also be about systems and infrastructure, such as administrative records about businesses and public services. Data is increasingly used to describe location, such as geospatial reference details, and the environment we live in, such as data about biodiversity or the weather. It can also refer to the information generated by the burgeoning web of sensors that make up the Internet of Things.”

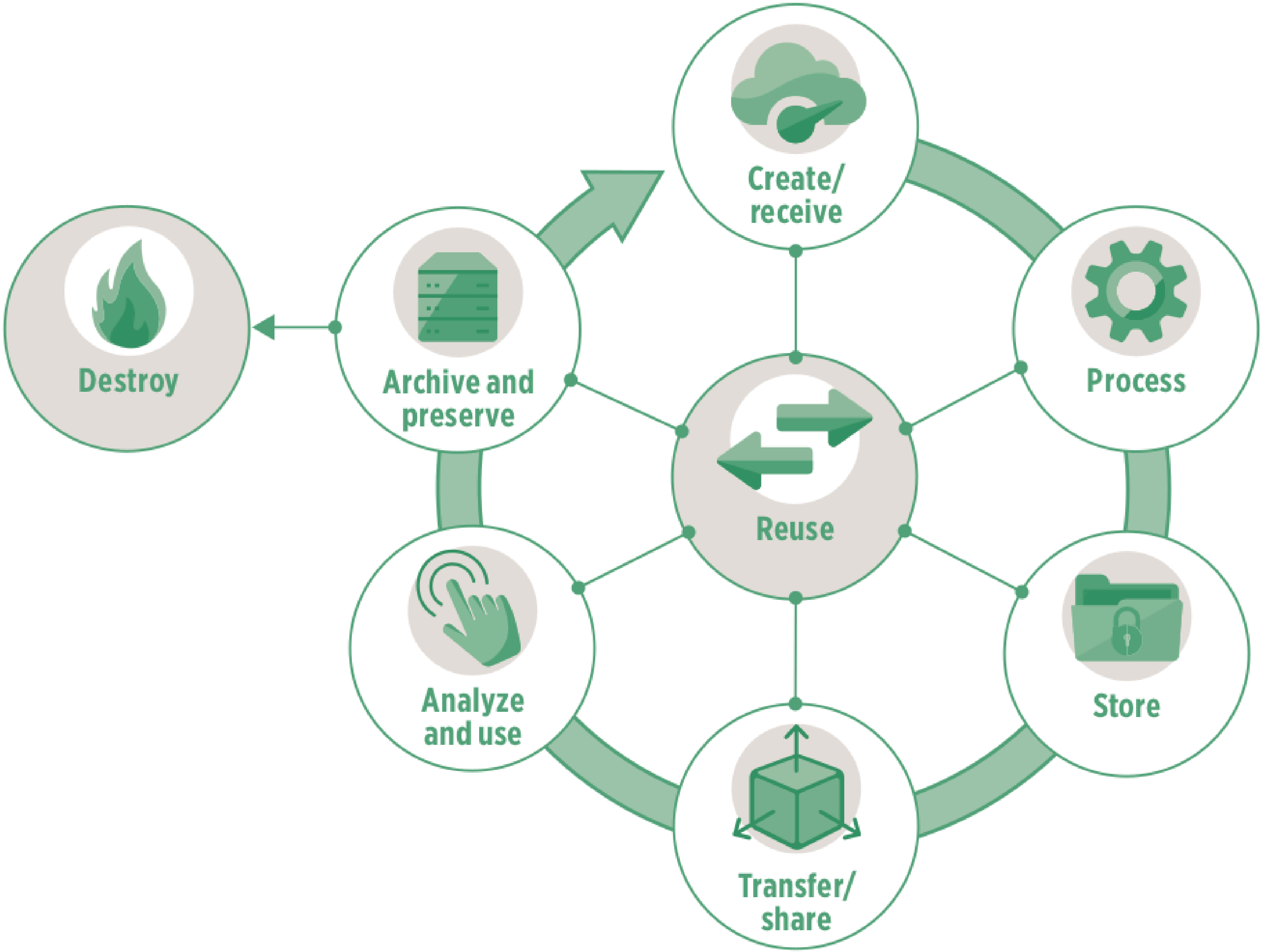
- UK National Data Strategy

What is data governance?

A system of **robust legal frameworks**, **ethical guidelines**, **standards**, and **processes** that ensures an organisation’s data is **accurate**, **consistent**, **secure**, and **compliant** throughout its lifecycle.



Data lifecycle



Public intent data

- Public intent data is data collected for public purposes.
- Public intent data is a foundation of public policies and can play a transformative role in the public sector.

Private intent data

- Private intent data is data collected by the private sector as part of routine business processes.
- Private intent data can drive innovation, lower transaction costs, and improve productivity, competitiveness, and economic growth.

Types of public intent data

- Administrative data
- Census data
- Sample surveys
- Citizen-generated data
- Machine-generated data
- Geospatial data

Maximising value of public intent data

Ensuring the data have adequate coverage	Ensuring the data are of high quality	Ensuring the data are easy to use	Ensuring the data are safe to use
• Completeness • Timeliness • Frequency	• Granularity • Accuracy • Comparability	• Accessibility • Understandability • Interoperability	• Impartiality • Confidentiality • Appropriateness

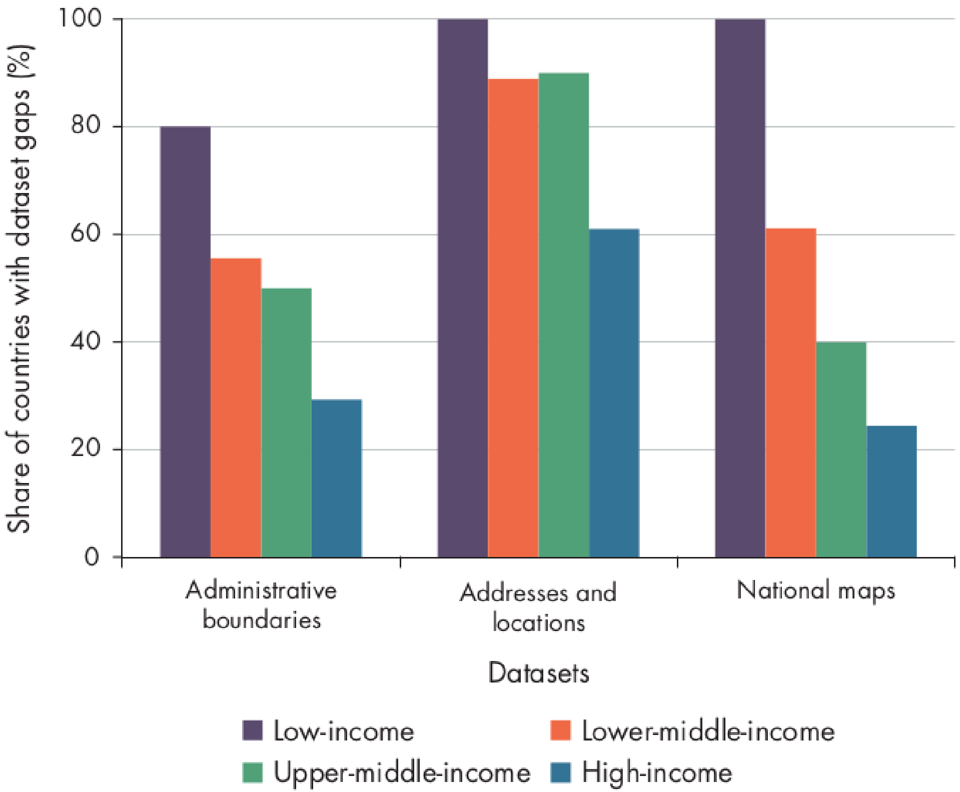
Three pathways for public intent data to add value

- Improving service delivery
- Prioritising scarce resources
- Holding government accountable and empowering individuals

Gaps in public intent data

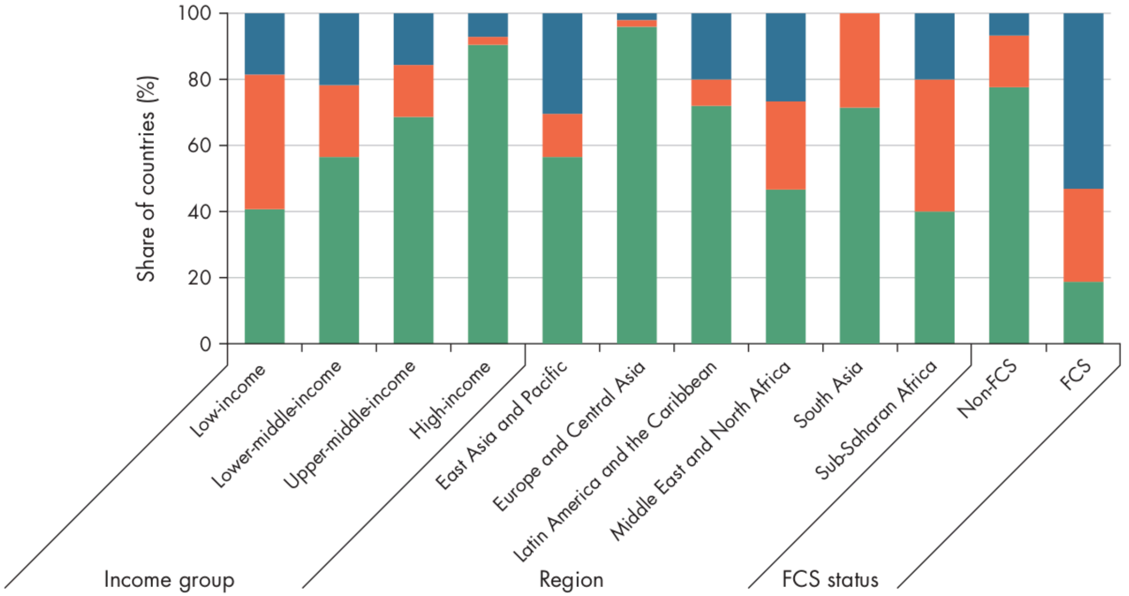
Coverage is inadequate

- Lack of *timeliness*, *frequency*, and *completeness*.
- Large gaps in geospatial data in lower-income countries



Data quality is poor

- Lack of *granularity*, *accuracy*, and *comparability*.
- Lack of data disaggregation
- Lower-income countries have less comparable poverty data



Data not easy to use

- Lack of *accessibility*, *understandability*, and *interoperability*.

Indicator	Low-income	Low-middle-income	Upper-middle-income	High-income
Openness score (0–100)	38	47	50	66
Available in machine readable format (%)	37	51	61	81
Available in non-proprietary format (%)	75	85	81	84
Download options available (%)	56	68	68	78
Open terms of use/license (%)	11	19	22	44

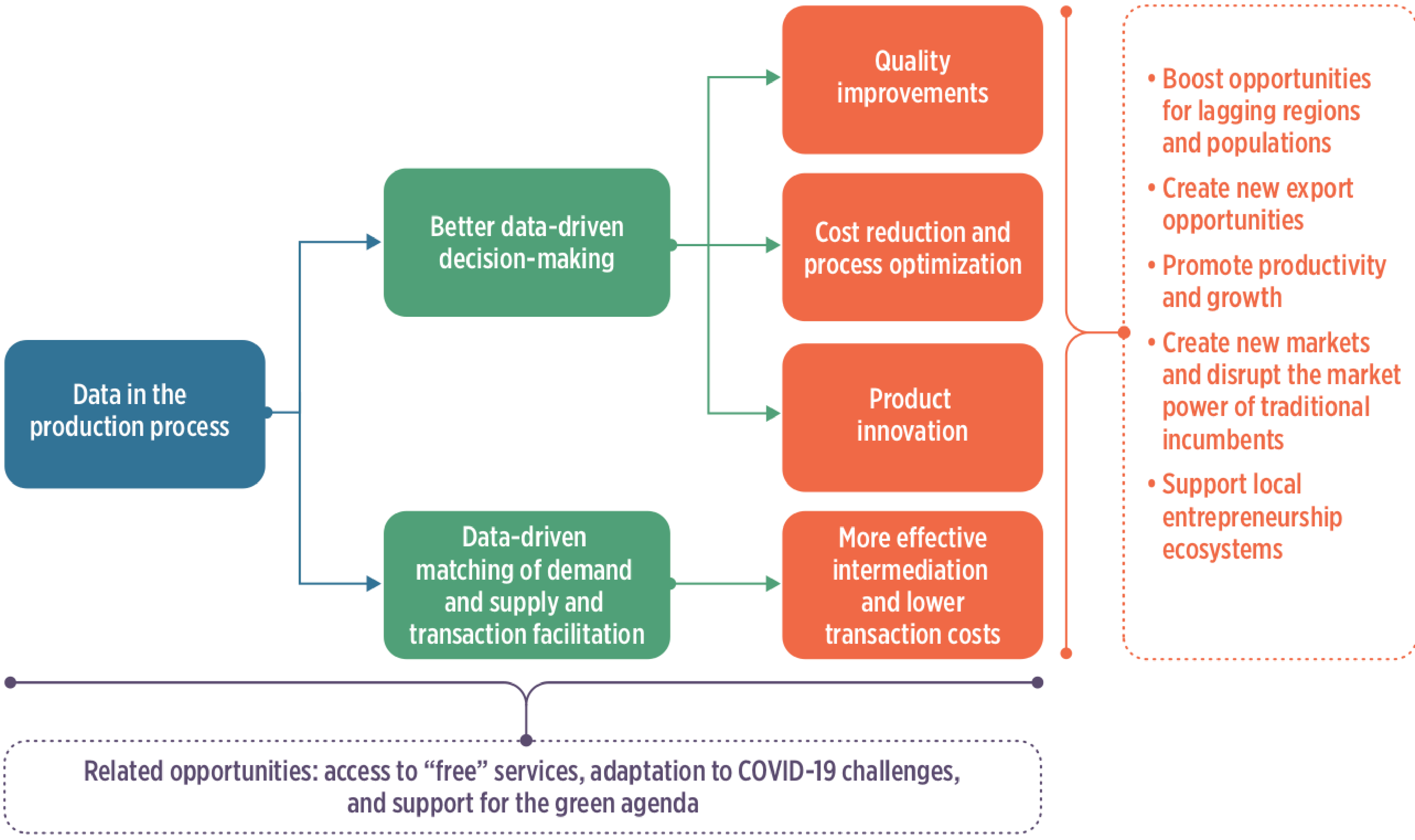
Data not safe to use

- Data not immune to influence from stakeholders
- Data not stored securely - too common in government and private sector databases
- Data not properly de-identified

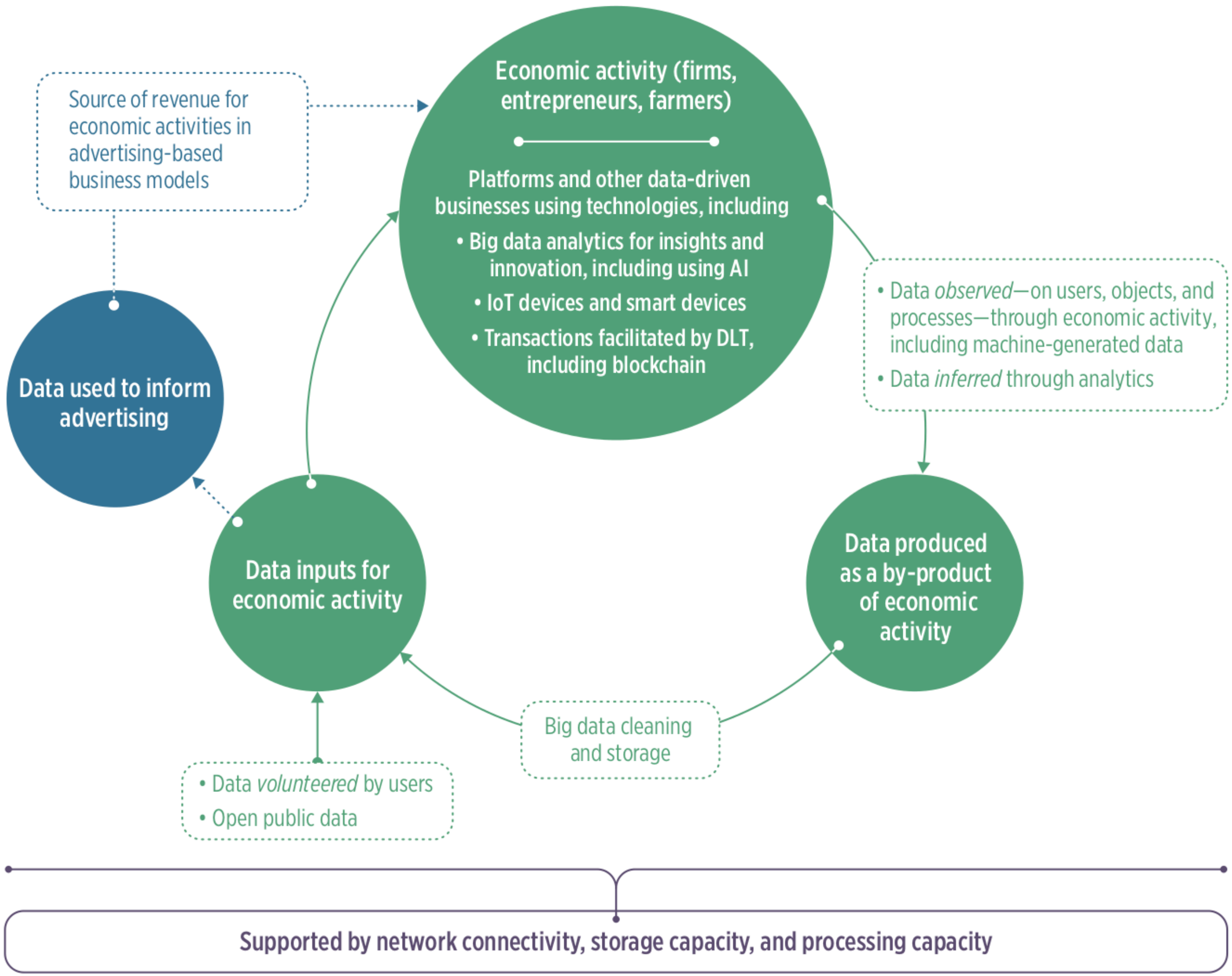
Why gaps persist

- Deficiencies in financing
- Deficiencies in technical capacity
- Deficiencies in governance
- Deficiencies in data demand

Data in the production process



Role of data in economic activity



Data inputs for economic activity

- Data collection by firms - **“digital footprint”**
- Use of public intent data by businesses

Sectors transformed by use of data in the production process

- Finance
- Agriculture
- Health
- Education
- Transport and logistics

Finance sector

- **Alternative credit scoring algorithms** - use of an individual's digital footprint to train machine learning algorithms to identify, score, and underwrite credit.
- **Payment systems** - digital payments; mobile payments.
- **Use of transaction data for product development** - Mastercard's Tourism Insights service allows leveraging big data to provide information on travellers' preferences.
- **Distributed ledger technology** - blockchain; eliminates financial intermediaries; embedding rules into smart contracts simplifies negotiation and verification processes.

Agriculture sector

- Remote sensing and geographic information systems - analysis of these data provide insights into farming operations; smart farming
- Internet-of-things (IoT) and embedded devices for monitoring farm and livestock conditions and farm operations
- Use of data to provide a range of services along the agriculture value chain
- Use of data to improve food traceability

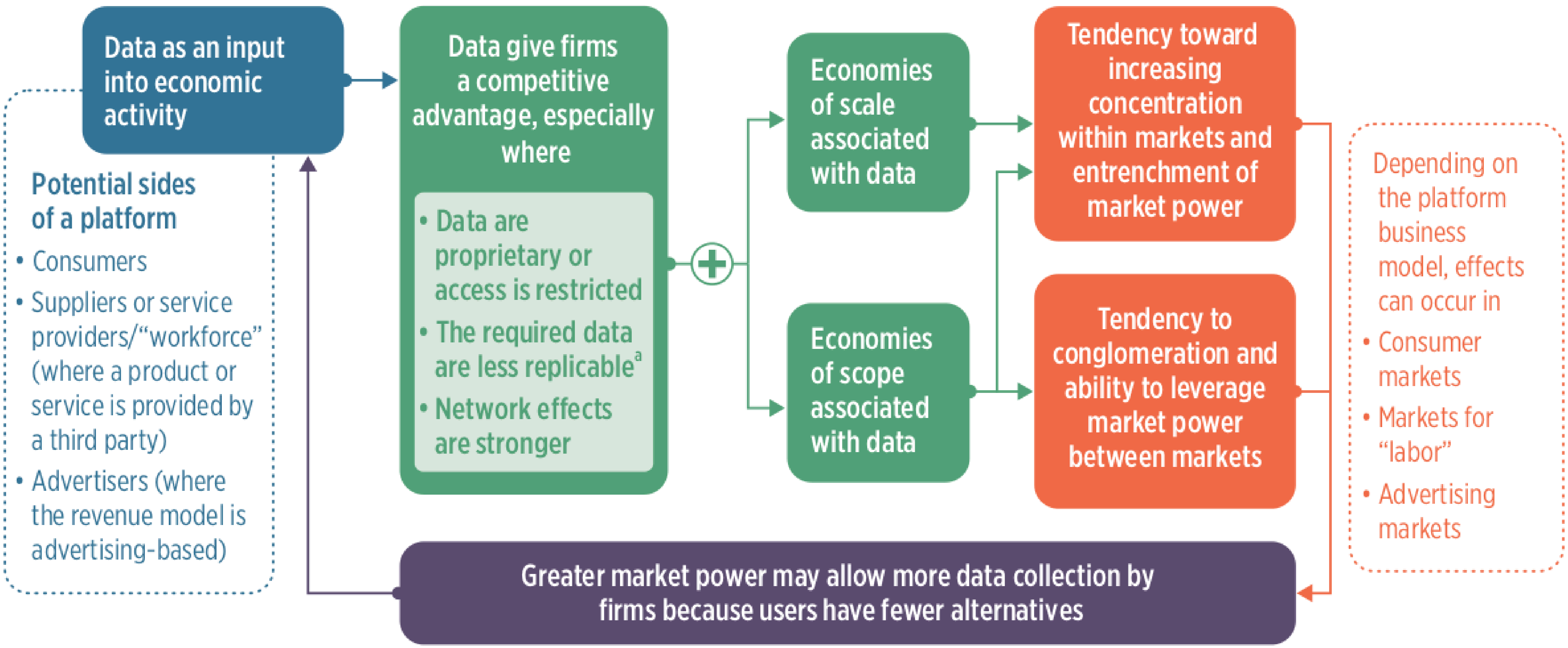
Education

- video conferencing/video communication tools
- digital supplemental learning platforms
- intelligent adaptive education

Transport and logistics sector

- Digital freight matching
- Digital courier logistics
- IoT-enabled cold chain

Risks to market structure from platform firms



Risks and adverse outcomes of data-driven businesses

- Increase propensity of dominant firms emerging
- Potential for exploitation of individuals
- Potential to increase inequality within and among countries